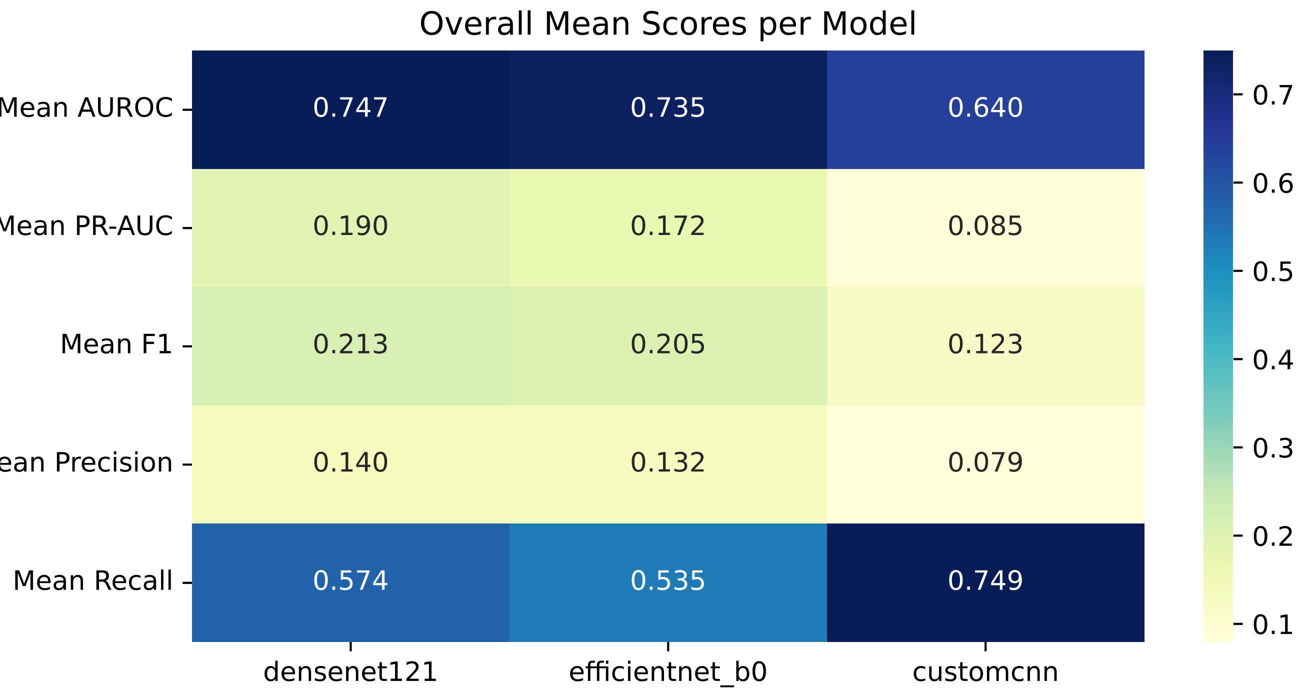
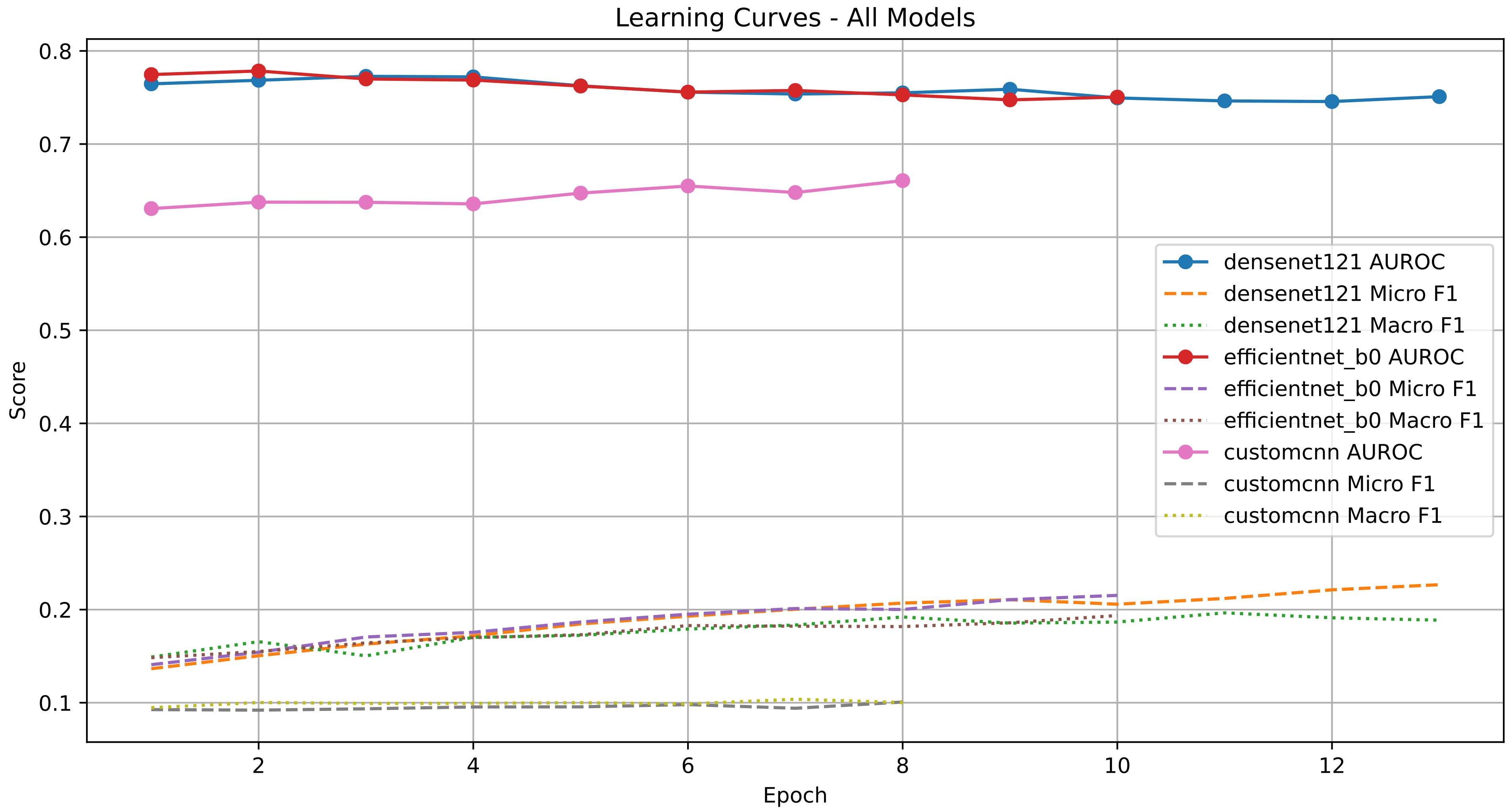
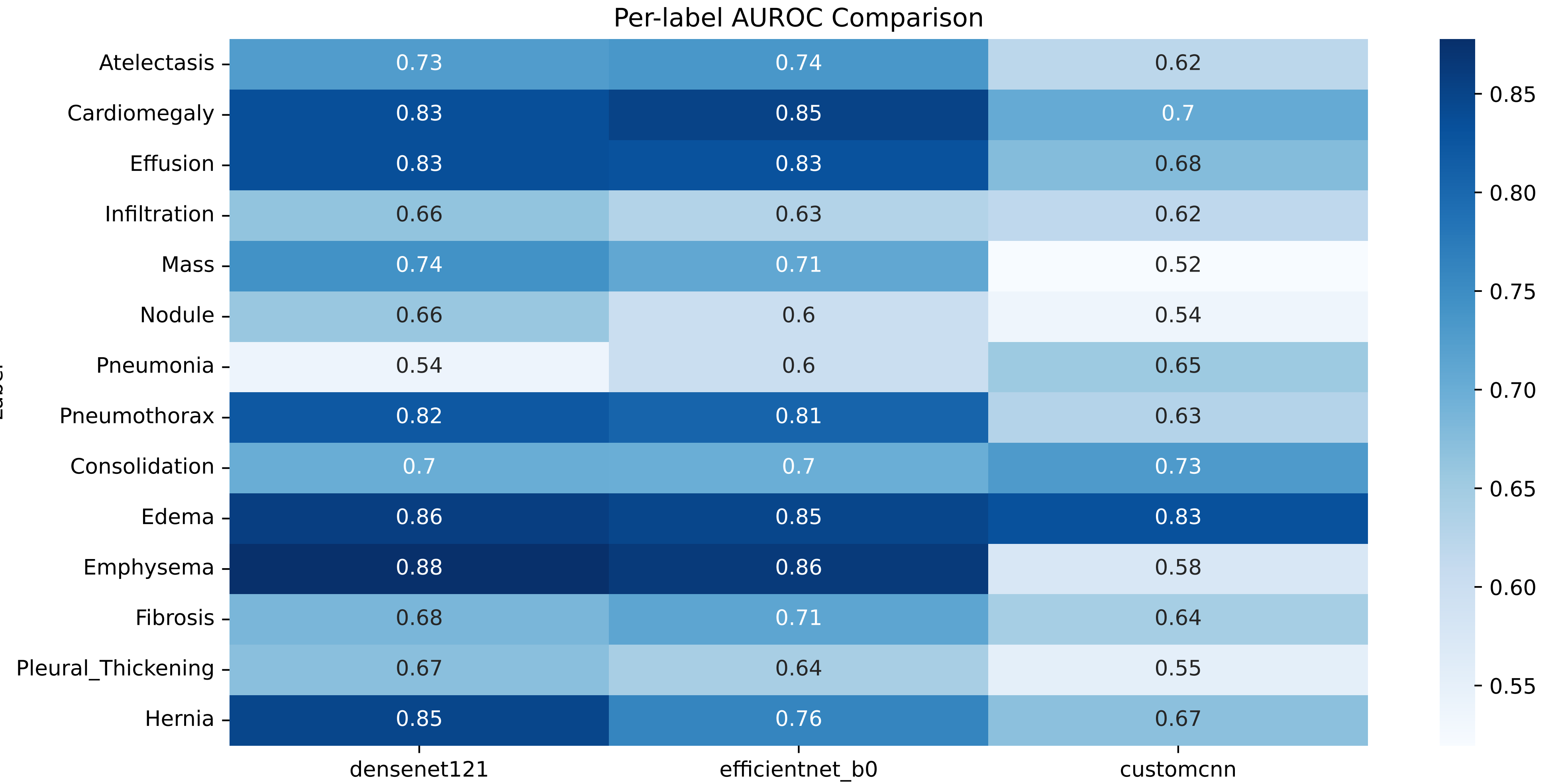
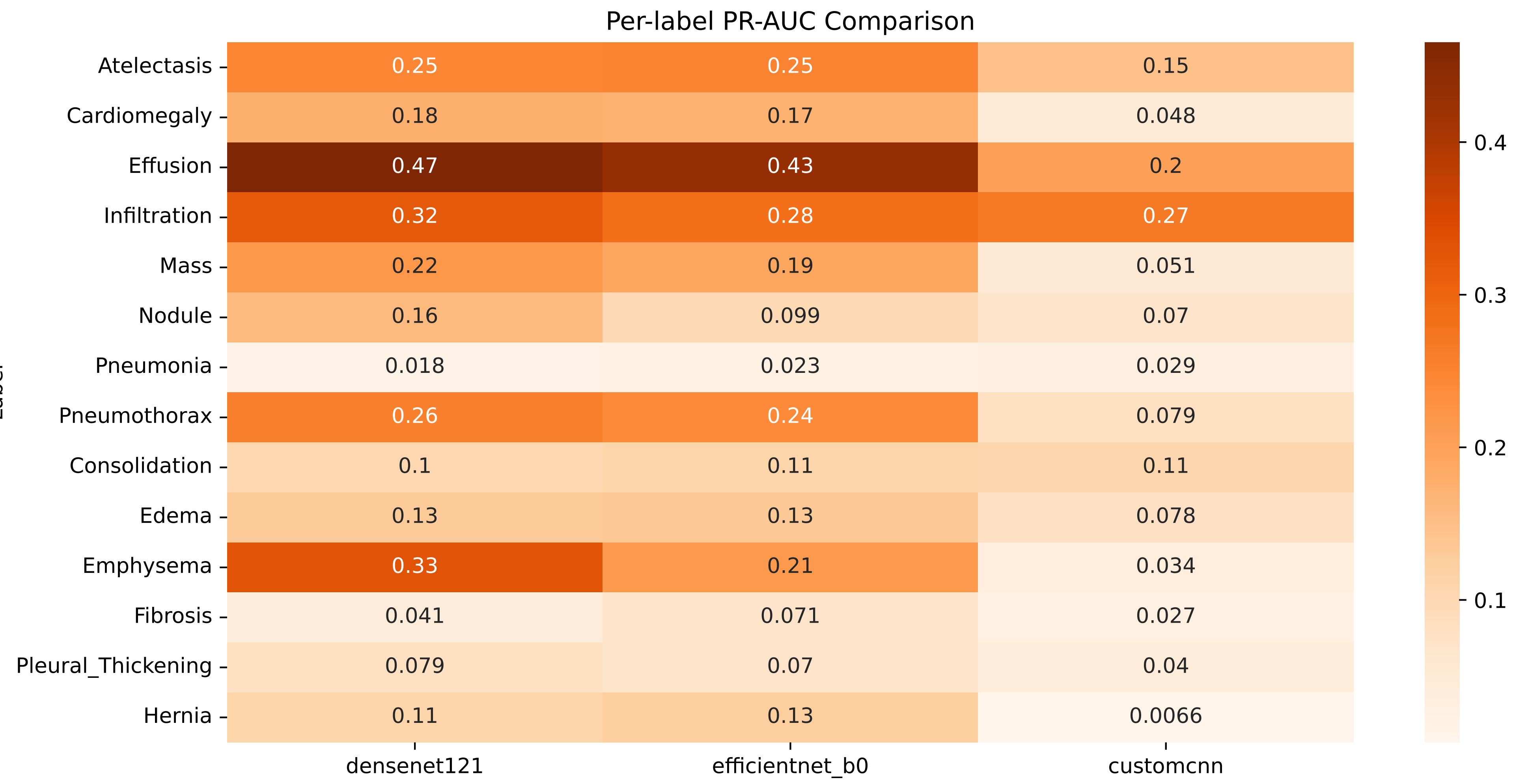
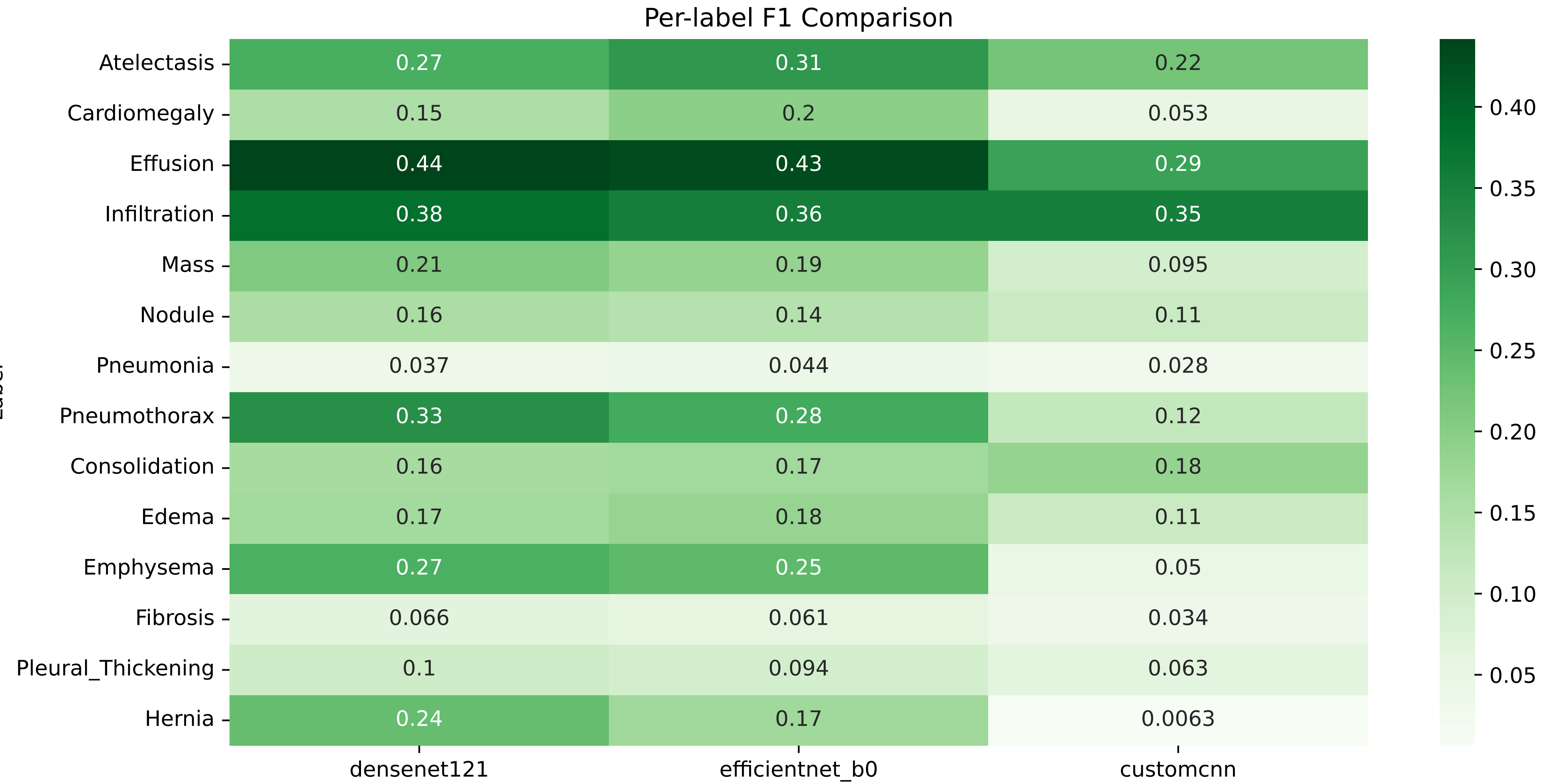
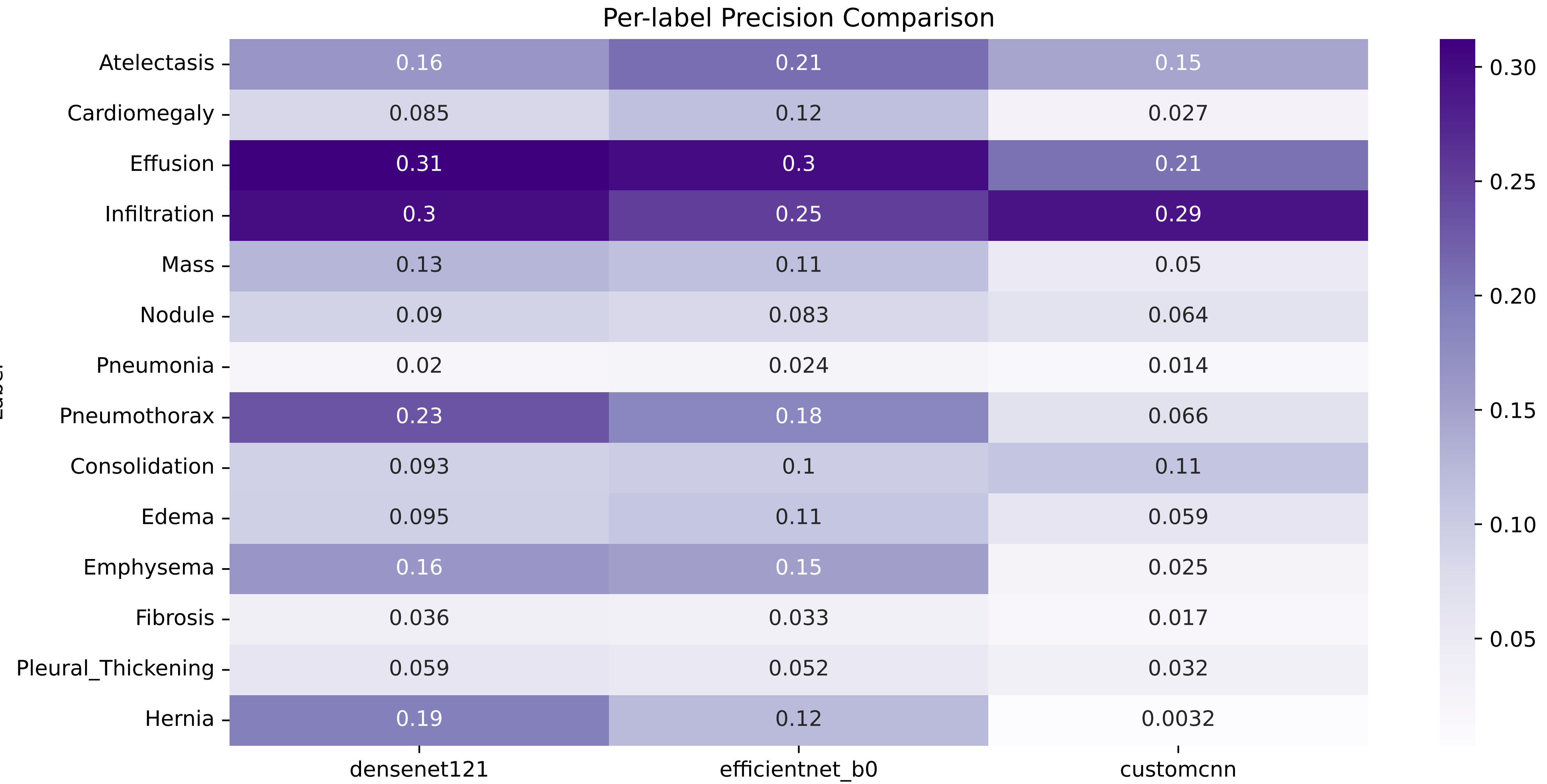

Overall Mean Scores per Model
Mean AUROC
Mean PR-AUC
Mean F1
ean Precision
Mean Recall
densenet121
efficientnet_b0
customcnn
0.747
0.735
0.640
0.190
0.172
0.085
0.213
0.205
0.123
0.140
0.132
0.079
0.574
0.535
0.749
0.7
0.6
0.5
0.4
0.3
0.2
0.1

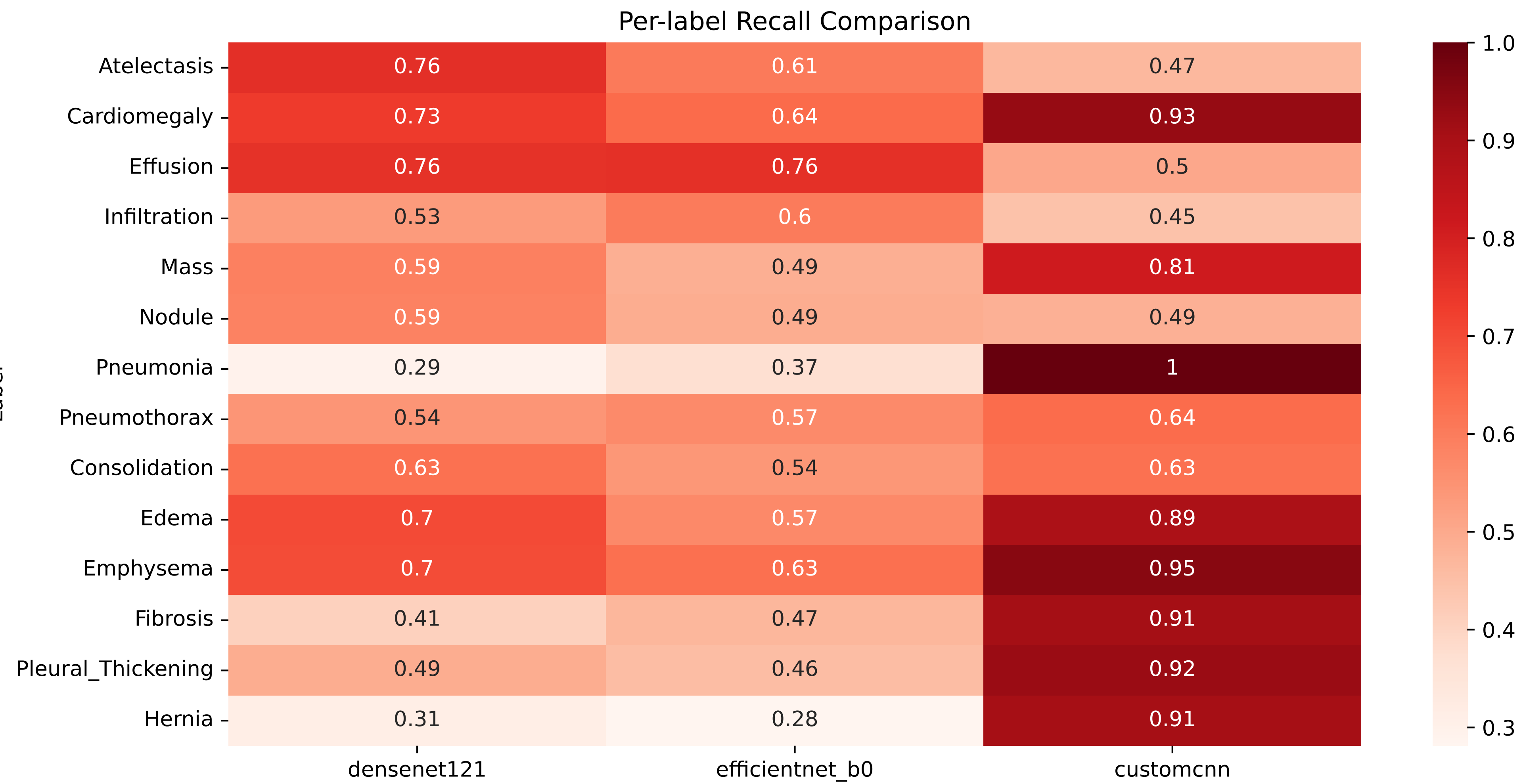

Learning Curves - All Models
Score
Epoch
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
2
4
6
8
10
12
densenet121 AUROC
densenet121 Micro F1
densenet121 Macro F1
efficientnet_b0 AUROC
efficientnet_b0 Micro F1
efficientnet_b0 Macro F1
customcnn AUROC
customcnn Micro F1
customcnn Macro F1


Per-label AUROC Comparison
Atelectasis
Cardiomegaly
Effusion
Infiltration
Mass
Nodule
Pneumonia
Pneumothorax
Consolidation
Edema
Emphysema
Fibrosis
Pleural_Thickening
Hernia
densenet121
efficientnet_b0
customcnn
0.73 0.74 0.62
0.83 0.85 0.7
0.83 0.83 0.68
0.66 0.63 0.62
0.74 0.71 0.52
0.66 0.6 0.54
0.54 0.6 0.65
0.82 0.81 0.63
0.7 0.7 0.73
0.86 0.85 0.83
0.88 0.86 0.58
0.68 0.71 0.64
0.67 0.64 0.55
0.85 0.76 0.67
0.85
0.80
0.75
0.70
0.65
0.60
0.55

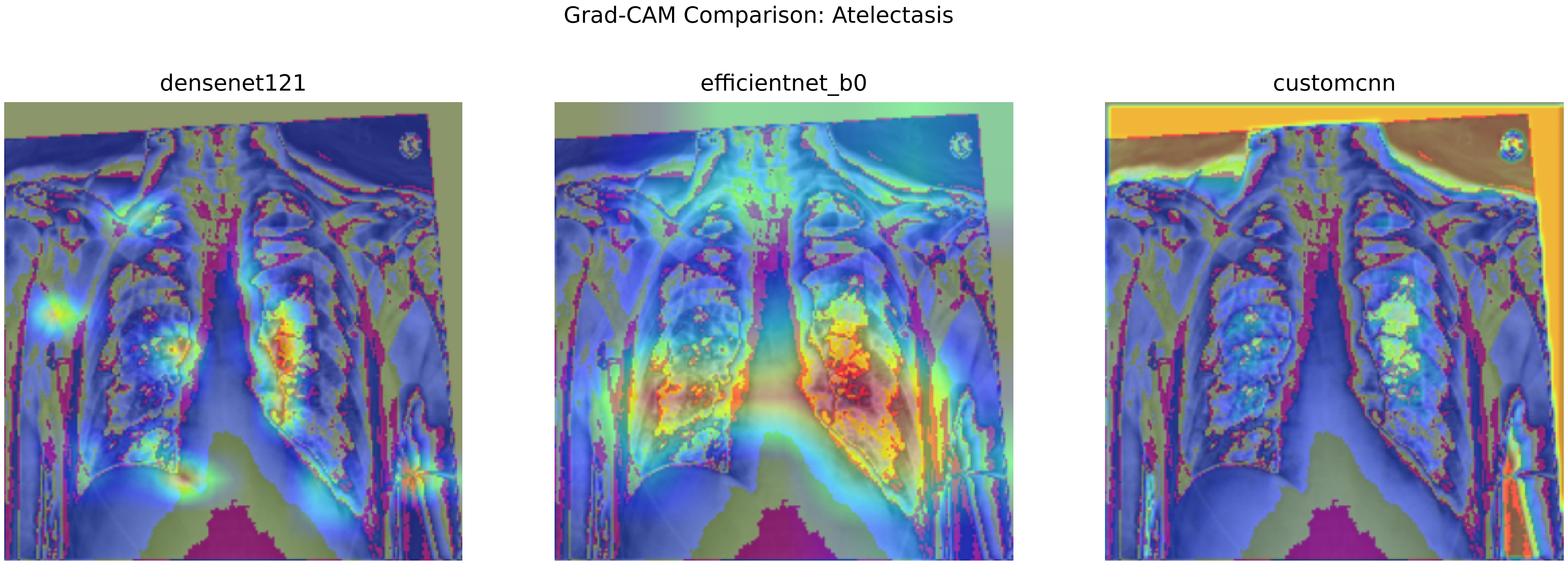

Per-label PR-AUC Comparison
Atelectasis
Cardiomegaly
Effusion
Infiltration
Mass
Nodule
Pneumonia
Pneumothorax
Consolidation
Edema
Emphysema
Fibrosis
Pleural_Thickening
Hernia
densenet121
efficientnet_b0
customcnn
0.25 0.25 0.15
0.18 0.17 0.048
0.47 0.43 0.2
0.32 0.28 0.27
0.22 0.19 0.051
0.16 0.099 0.07
0.018 0.023 0.029
0.26 0.24 0.079
0.1 0.11 0.11
0.13 0.13 0.078
0.33 0.21 0.034
0.041 0.071 0.027
0.079 0.07 0.04
0.11 0.13 0.0066
0.4
0.3
0.2
0.1


Per-label F1 Comparison
Label
Atelectasis
Cardiomegaly
Effusion
Infiltration
Mass
Nodule
Pneumonia
Pneumothorax
Consolidation
Edema
Emphysema
Fibrosis
Pleural_Thickening
Hernia
densenet121
efficientnet_b0
customcnn
0.27 0.31 0.22
0.15 0.2 0.053
0.44 0.43 0.29
0.38 0.36 0.35
0.21 0.19 0.095
0.16 0.14 0.11
0.037 0.044 0.028
0.33 0.28 0.12
0.16 0.17 0.18
0.17 0.18 0.11
0.27 0.25 0.05
0.066 0.061 0.034
0.1 0.094 0.063
0.24 0.17 0.0063
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05


Per-label Precision Comparison
Atelectasis
Cardiomegaly
Effusion
Infiltration
Mass
Nodule
Pneumonia
Pneumothorax
Consolidation
Edema
Emphysema
Fibrosis
Pleural_Thickening
Hernia
densenet121
efficientnet_b0
customcnn
0.16 0.21 0.15
0.085 0.12 0.027
0.31 0.3 0.21
0.3 0.25 0.29
0.13 0.11 0.05
0.09 0.083 0.064
0.02 0.024 0.014
0.23 0.18 0.066
0.093 0.1 0.11
0.095 0.11 0.059
0.16 0.15 0.025
0.036 0.033 0.017
0.059 0.052 0.032
0.19 0.12 0.0032
0.30
0.25
0.20
0.15
0.10
0.05



Per-label Recall Comparison
Atelectasis
Cardiomegaly
Effusion
Infiltration
Mass
Nodule
Pneumonia
Pneumothorax
Consolidation
Edema
Emphysema
Fibrosis
Pleural_Thickening
Hernia
densenet121
efficientnet_b0
customcnn
0.76
0.61
0.47
0.73
0.64
0.93
0.76
0.76
0.5
0.53
0.6
0.45
0.59
0.49
0.81
0.59
0.49
0.49
0.29
0.37
1
0.54
0.57
0.64
0.63
0.54
0.63
0.7
0.57
0.89
0.7
0.63
0.95
0.41
0.47
0.91
0.49
0.46
0.92
0.31
0.28
0.91
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3


Grad-CAM Comparison: Atelectasis
densenet121
efficientnet_b0
customcnn

Atelectasis LLM Summary

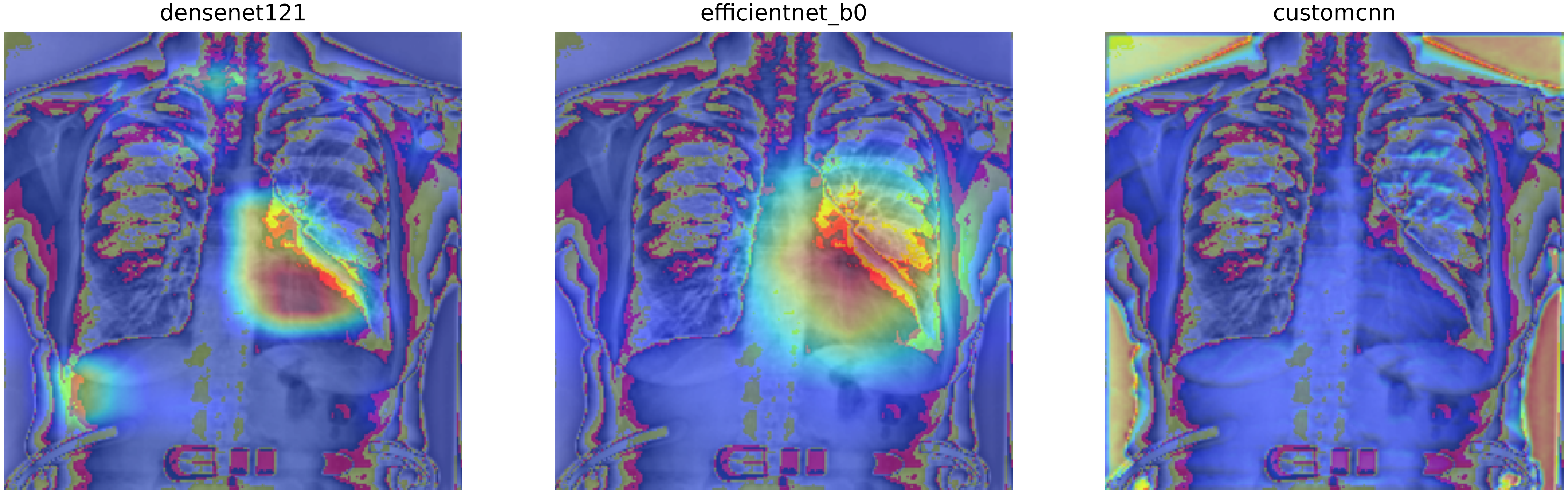
Based on the provided logs, the performance of the models on **Atelectasis** is as follows:

*   **EfficientNet-B0:** Achieved an AUROC of 0.736, PR-AUC of 0.251, and F1-score of 0.311.
*   **DenseNet121:** Achieved an AUROC of 0.727, PR-AUC of 0.246, and F1-score of 0.271.
*   **CustomCNN:** No performance data for this model was provided in the logs.

Comparison to ChestXray14 Benchmarks:
The performance of these models on Atelectasis (AUROC ~0.73) is lower than the benchmarks reported by Wang et al. (2017) for other conditions in the ChestXray14 dataset, such as Edema (AUROC ~0.84) and Cardiomegaly (AUROC ~0.81).

Citation:
*   Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

Grad-CAM Comparison: Cardiomegaly



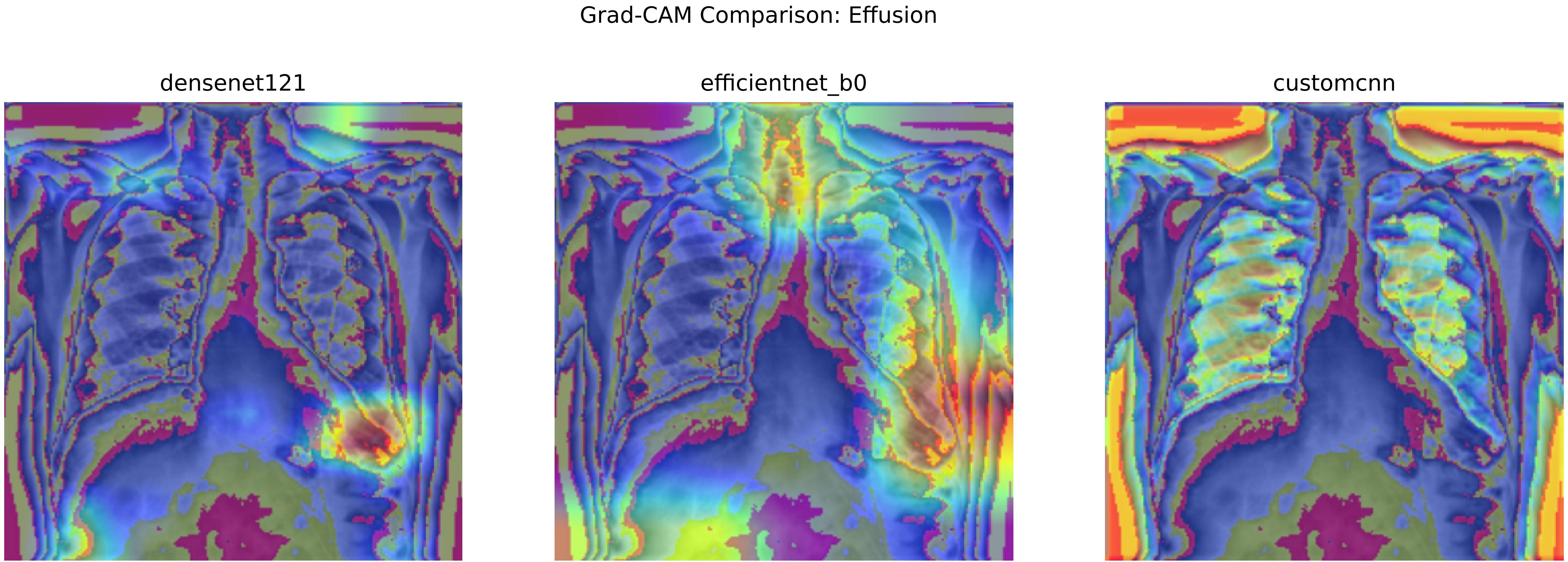
Cardiomegaly LLM Summary

Based on the provided logs, the performance of the three models on the Cardiomegaly task is as follows:

*   **EfficientNet-B0:** Achieved the highest performance with an AUROC of 0.851, a PR-AUC of 0.173, and an F1 score of 0.195.
*   **DenseNet121:** Followed closely with an AUROC of 0.834, a PR-AUC of 0.177, and an F1 score of 0.152.
*   **CustomCNN:** Significantly underperformed compared to the other two architectures, recording an AUROC of 0.705, a PR-AUC of 0.048, and an F1 score of 0.053.

Comparison to Benchmarks:
When compared to the ChestXray14 benchmarks established by Wang et al. (2017), both EfficientNet-B0 and DenseNet121 demonstrate competitive results. While Wang et al. (2017) reported an AUROC of approximately 0.81 for Cardiomegaly, both EfficientNet-B0 and DenseNet121 exceeded this benchmark, whereas the CustomCNN fell below it.

Reference:
Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.


Grad-CAM Comparison: Effusion
densenet121
efficientnet_b0
customcnn

Effusion LLM Summary

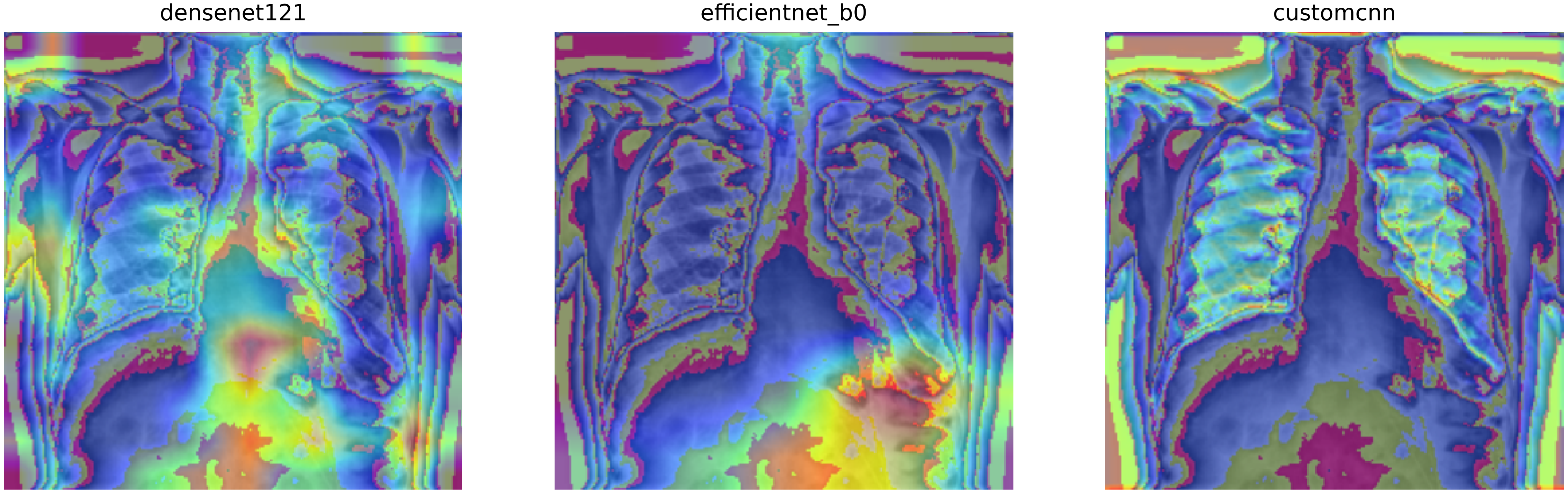
Based on the provided logs, the performance of the models on the **Effusion** pathology is as follows:

*   **DenseNet121:** Achieved the highest performance with an AUROC of 0.835, PR-AUC of 0.465, and F1-score of 0.442.
*   **EfficientNet-B0:** Recorded an AUROC of 0.830, PR-AUC of 0.432, and F1-score of 0.431.
*   **CustomCNN:** Performance data for this model was not provided in the logs.

Comparison to ChestXray14 Benchmarks:
The performance of DenseNet121 and EfficientNet-B0 on Effusion (AUROC ~0.83) is competitive with the benchmarks established by Wang et al. (2017) for other common pathologies in the ChestXray14 dataset, such as Edema (AUROC ~0.84) and Cardiomegaly (AUROC ~0.81).

Citation:
*   Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Grad-CAM Comparison: Infiltration



Infiltration LLM Summary

Based on the provided logs, the performance of the models on the **Infiltration** pathology is as follows:

*   **DenseNet121:** Achieved the highest performance among the tested models with an AUROC of 0.665, PR-AUC of 0.321, and F1-score of 0.382.
*   **EfficientNet-B0:** Recorded an AUROC of 0.631, PR-AUC of 0.283, and F1-score of 0.356.
*   **CustomCNN:** Performance data for this model was not provided in the logs.

Comparison to ChestXray14 Benchmarks
When compared to the original ChestXray14 benchmarks established by Wang et al. (2017), where the reported AUROC for Infiltration was approximately 0.67, the **DenseNet121** model in our logs performs competitively, nearly matching the baseline. The **EfficientNet-B0** model shows slightly lower performance than the established benchmark.

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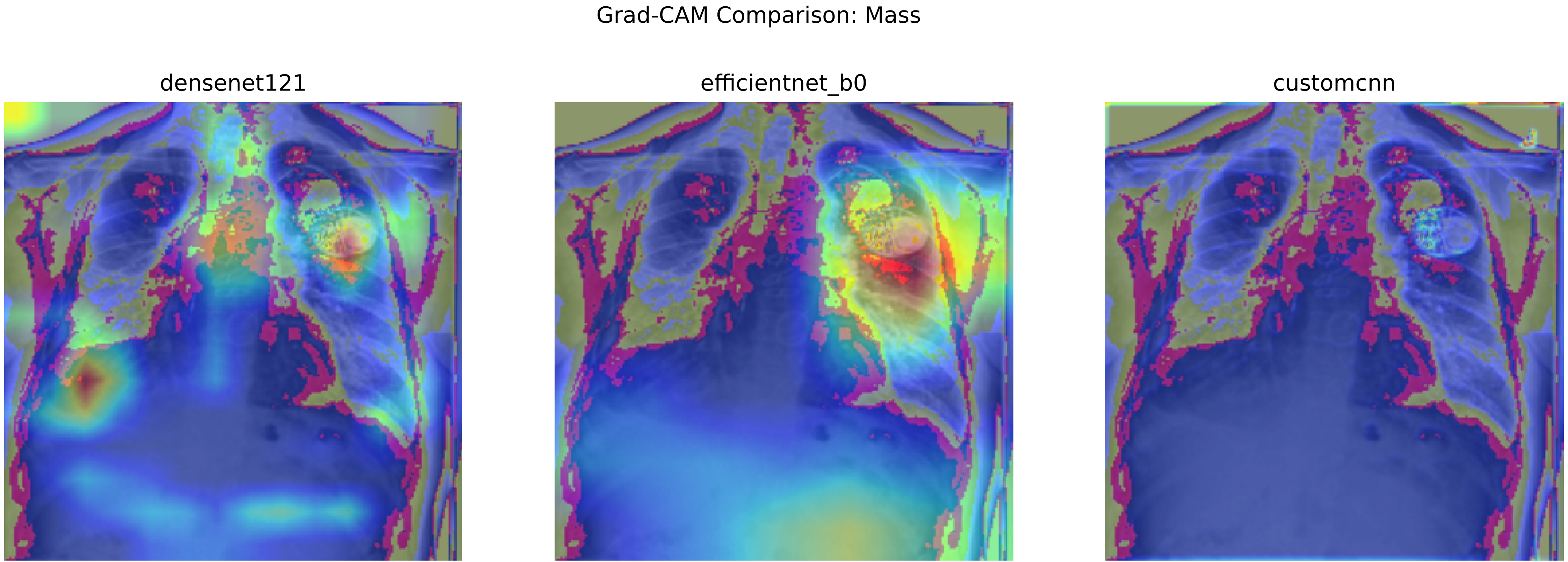
Citation:
Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2097-2106.


Grad-CAM Comparison: Mass
densenet121
efficientnet_b0
customcnn

Mass LLM Summary

Based on the provided logs and benchmark data, here is the performance summary and comparison:

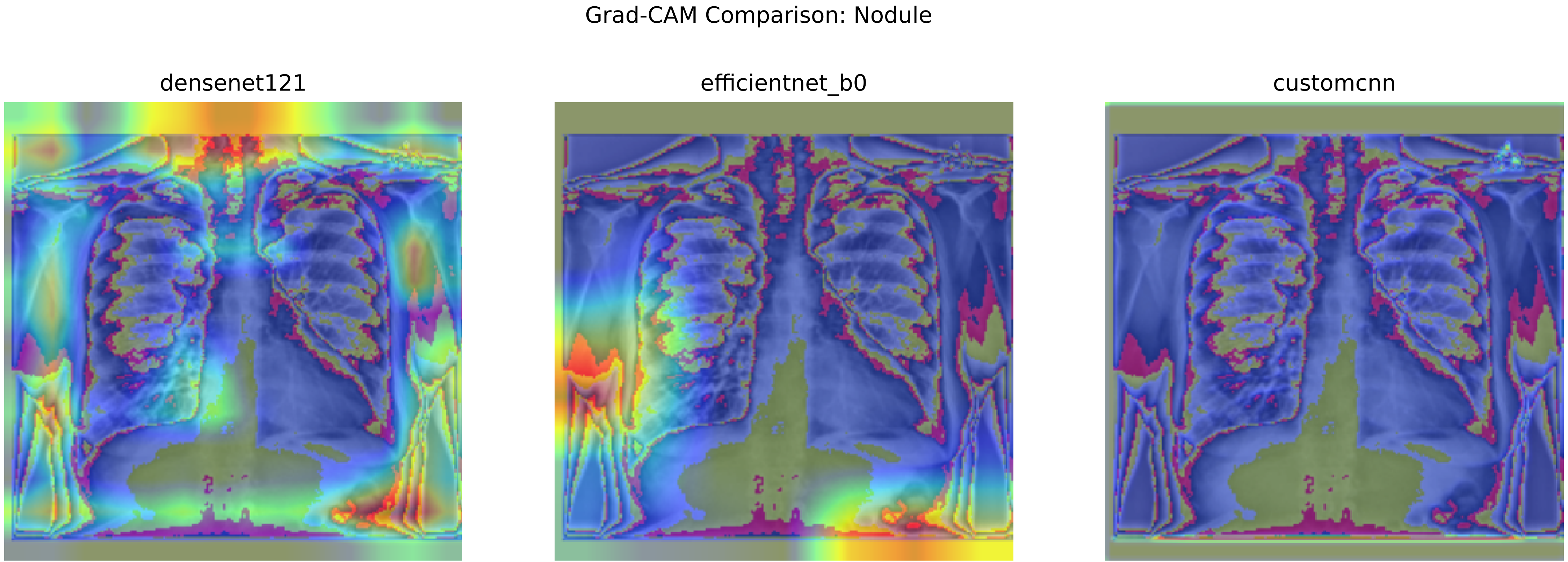
Performance Summary on Mass
The logs indicate that **EfficientNet-B0** achieved an AUROC of 0.709, a PR-AUC of 0.192, and an F1-score of 0.186 for the "Mass" classification task. (Note: The provided context does not contain performance data for DenseNet121 or CustomCNN specifically regarding the "Mass" category).

Comparison to ChestXray14 Benchmarks
When comparing these results to the benchmarks established by **Wang et al. (2017)**, the performance on Mass (AUROC 0.709) is notably lower than the benchmarks reported for other pathologies in the ChestXray14 dataset, such as Edema (AUROC ~0.84) and Cardiomegaly (AUROC ~0.81).

It is important to note that performance in these tasks is highly dependent on data distribution; for instance, Wang et al. (2017) highlight that performance for certain conditions, such as Hernia, remains low due to a scarcity of positive training samples, which may also be a factor in the performance metrics observed for Mass.

References:

*   Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.


Grad-CAM Comparison: Nodule
densenet121
efficientnet_b0
customcnn

Nodule LLM Summary

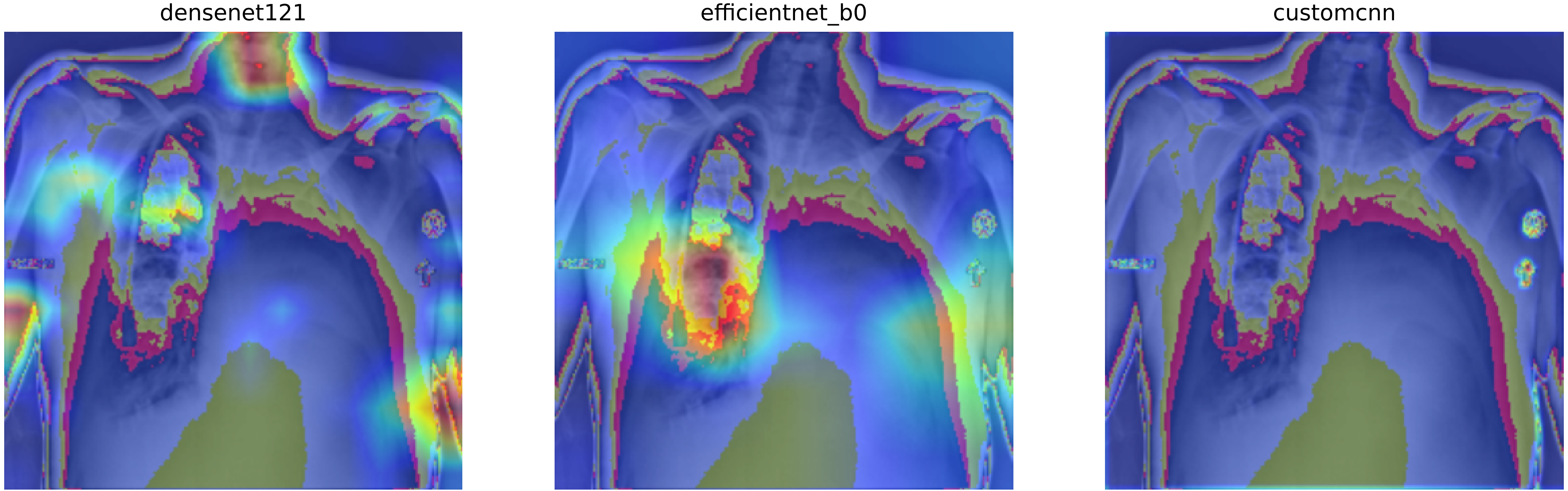
Based on the provided logs, the performance of the models on the **Nodule** classification task is as follows:

*   **DenseNet121:** Achieved the highest performance among the tested models with an AUROC of 0.659, a PR-AUC of 0.158, and an F1-score of 0.156.
*   **EfficientNet-B0:** Recorded an AUROC of 0.601, a PR-AUC of 0.099, and an F1-score of 0.142.
*   **CustomCNN:** Performance data for this model was not provided in the logs.

Comparison to ChestXray14 Benchmarks:
While the provided logs show DenseNet121 outperforming EfficientNet-B0 on the Nodule task, these results (AUROC 0.601–0.659) are generally lower than the established benchmarks for the ChestXray14 dataset. In the original ChestXray14 study, state-of-the-art models typically report Nodule AUROC scores ranging from 0.70 to 0.80, depending on the architecture and training configuration (Wang et al., 2017). The current results suggest that further optimization or fine-tuning may be required to reach parity with standard benchmarks.

Citation:
Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Grad-CAM Comparison: Pneumonia



Pneumonia LLM Summary

Based on the provided logs, the performance of the three models on the Pneumonia dataset is summarized below:

Performance Summary
*   **CustomCNN:** Achieved the highest AUROC (0.655) and PR-AUC (0.029), though it recorded the lowest F1 score (0.028).
*   **EfficientNet-B0:** Demonstrated moderate performance with an AUROC of 0.601, a PR-AUC of 0.023, and the highest F1 score (0.044) among the three models.
*   **DenseNet121:** Showed the lowest performance across all metrics for Pneumonia, with an AUROC of 0.539, a PR-AUC of 0.018, and an F1 score of 0.037.

Comparison to ChestXray14 Benchmarks
When compared to established benchmarks for the ChestXray14 dataset, these models demonstrate significantly lower predictive capability. State-of-the-art models typically report Pneumonia AUROC scores ranging from 0.70 to 0.80+ (Wang et al., 2017). The results from the provided logs—particularly the low PR-AUC and F1 scores—suggest that these models are struggling with the class imbalance inherent in the Pneumonia detection task compared to standard benchmark performance.

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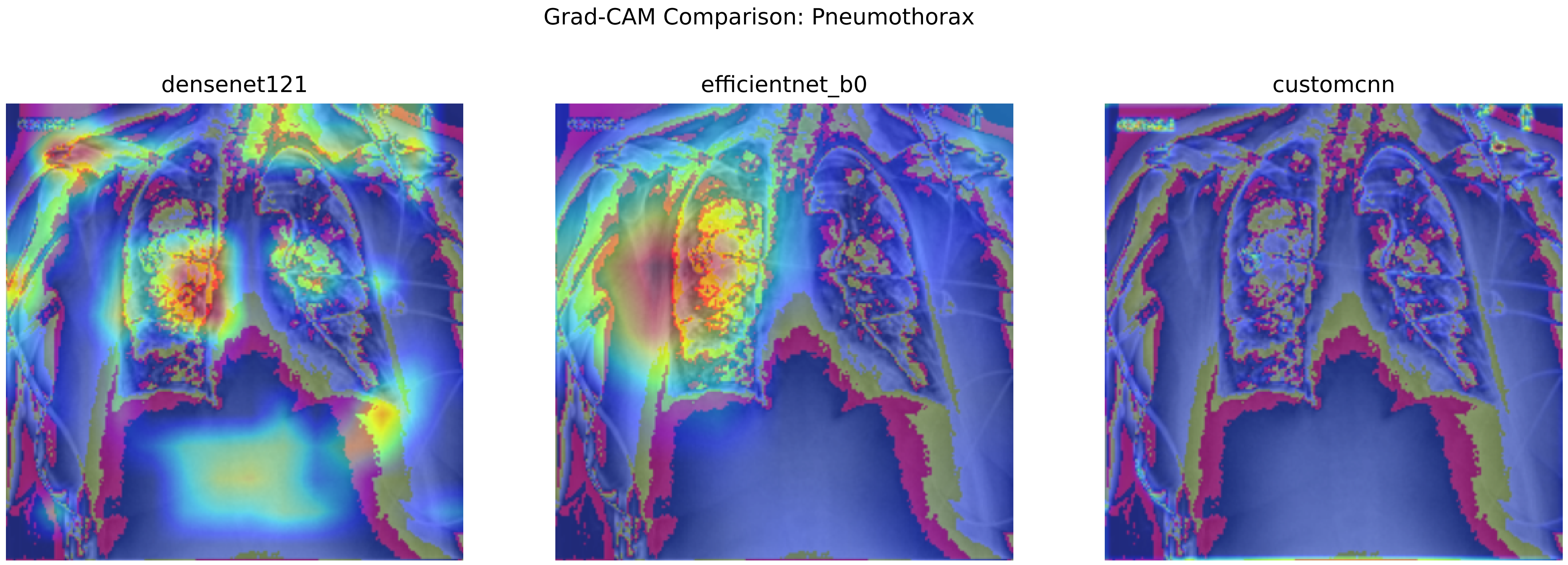
References:
*   Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2097-2106.


Grad-CAM Comparison: Pneumothorax
densenet121
efficientnet_b0
customcnn

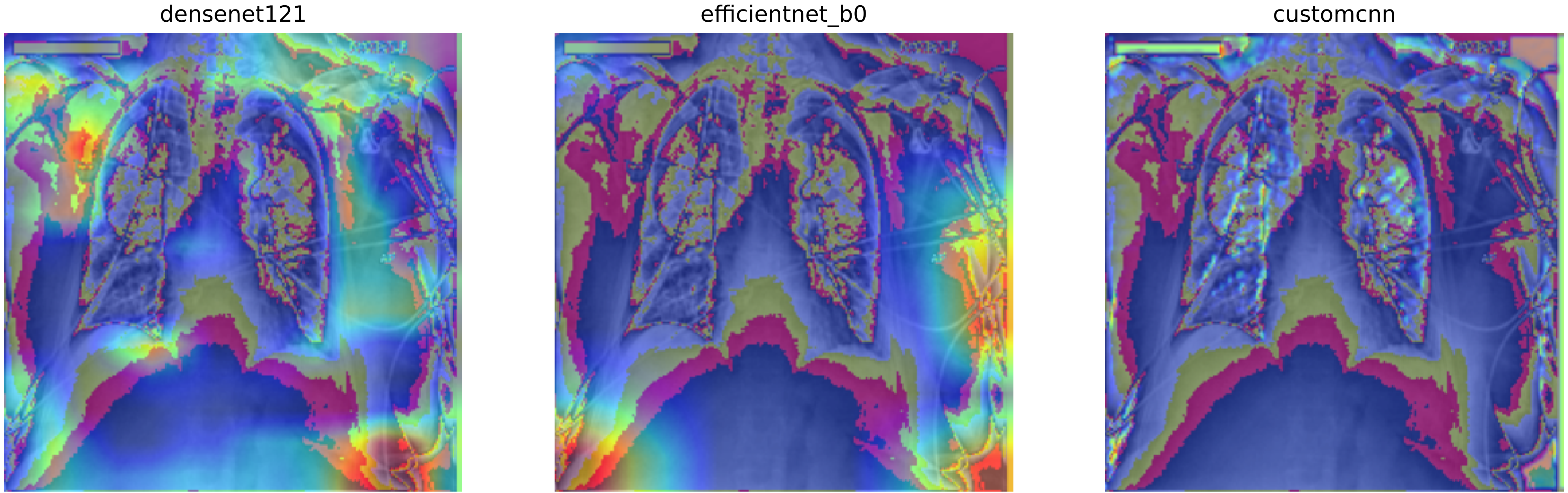
Pneumothorax LLM Summary

Based on the provided logs, the performance of the three models on the Pneumothorax dataset is as follows:

*   **DenseNet121** achieved the highest performance across all metrics, with an AUROC of 0.822, PR-AUC of 0.257, and F1-score of 0.325.
*   **EfficientNet-B0** followed closely, recording an AUROC of 0.806, PR-AUC of 0.240, and F1-score of 0.278.
*   **CustomCNN** significantly underperformed compared to the other two architectures, yielding an AUROC of 0.630, PR-AUC of 0.079, and F1-score of 0.119.

Comparison to ChestXray14 Benchmarks:
The performance of DenseNet121 and EfficientNet-B0 on Pneumothorax (AUROC ~0.81–0.82) is comparable to the benchmarks reported by Wang et al. (2017) for other pathologies in the ChestXray14 dataset, such as Cardiomegaly (AUROC ~0.81) and Edema (AUROC ~0.84). While these models show competitive results, it is worth noting that performance can vary significantly based on class prevalence, as evidenced by the low performance on Hernia due to a scarcity of positive samples (Wang et al., 2017).

Reference:
Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.



Grad-CAM Comparison: Consolidation

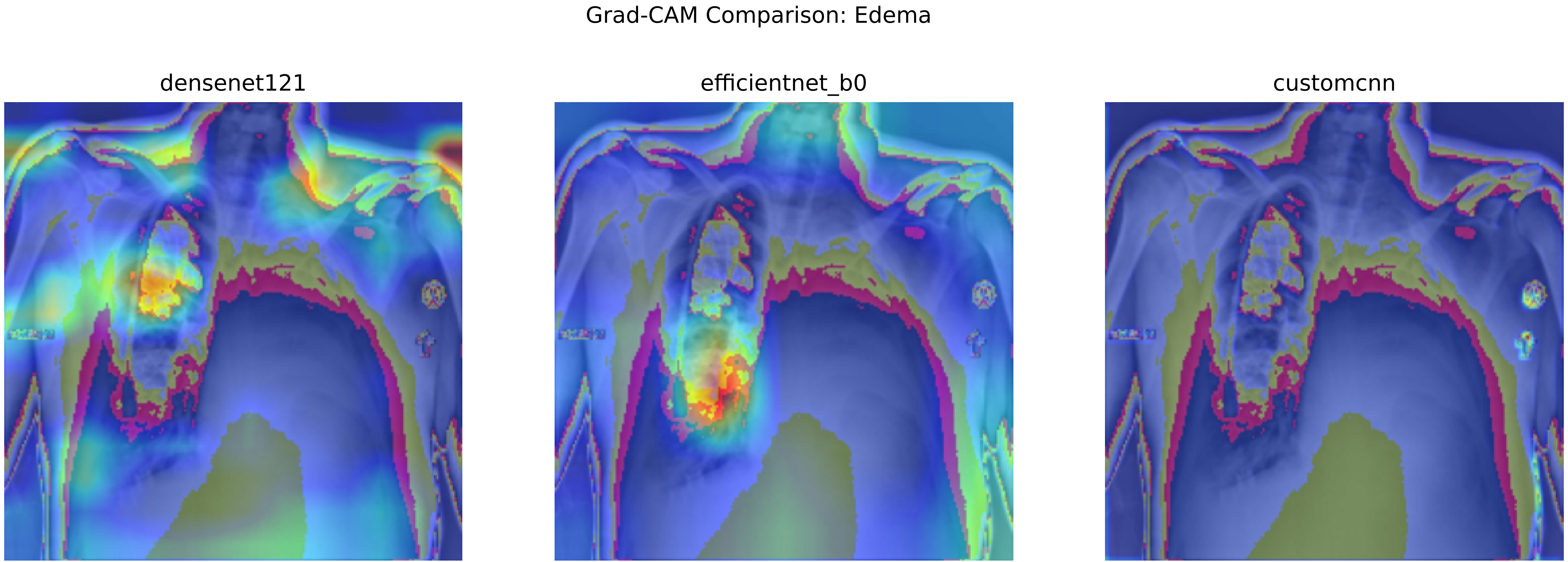
Consolidation LLM Summary

Based on the provided logs, the performance of the models on **Consolidation** is as follows:

*   **DenseNet121:** AUROC = 0.701, PR-AUC = 0.105, F1 = 0.162
*   **EfficientNet-B0:** AUROC = 0.699, PR-AUC = 0.109, F1 = 0.168
*   **CustomCNN:** Performance data for this model was not provided in the context.

Comparison to ChestXray14 Benchmarks:
The performance of both DenseNet121 and EfficientNet-B0 on Consolidation (AUROC ~0.70) is notably lower than the benchmarks reported by Wang et al. (2017) for other pathologies in the ChestXray14 dataset, such as Edema (AUROC ~0.84) and Cardiomegaly (AUROC ~0.81).

Citations:
*   Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.


Grad-CAM Comparison: Edema
densenet121
efficientnet_b0
customcnn

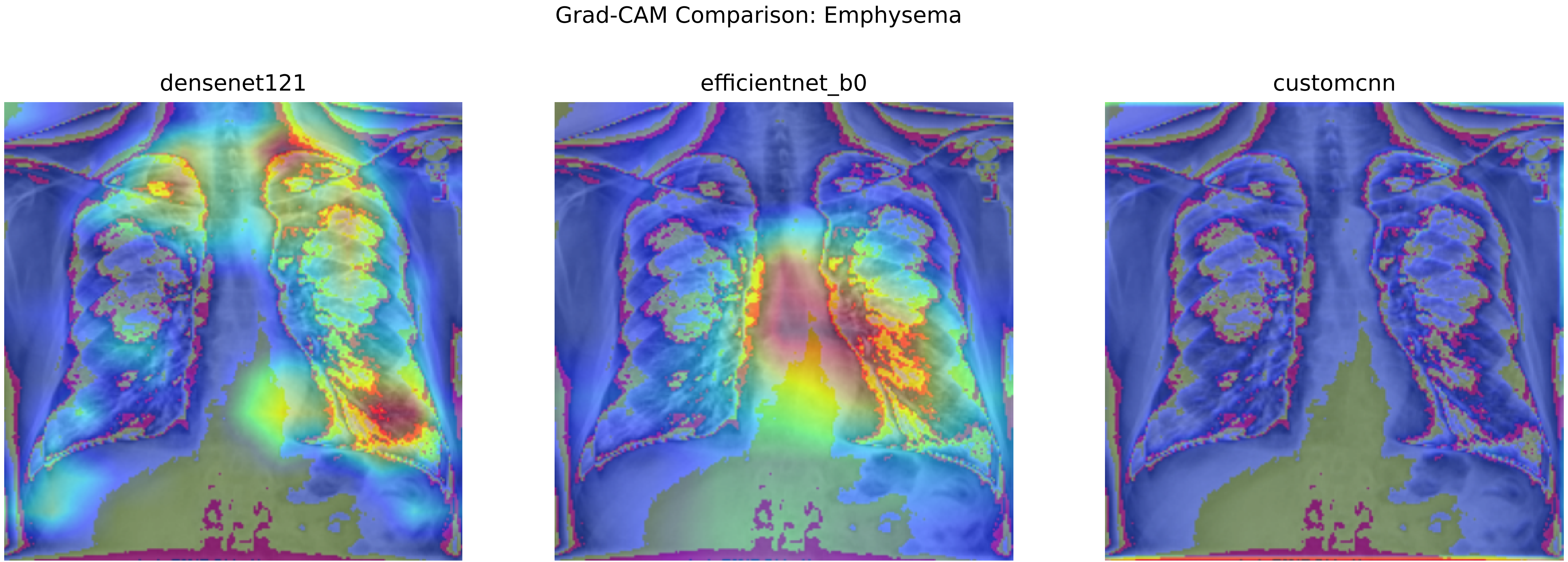
Edema LLM Summary

Based on the provided logs and the ChestXray14 benchmark, here is the performance summary for Edema detection:

Performance Summary
*   **DenseNet121:** Achieved the highest performance among the models tested with an **AUROC of 0.857** (PR-AUC=0.130, F1=0.167).
*   **EfficientNet-B0:** Performed closely to DenseNet121 with an **AUROC of 0.848** (PR-AUC=0.132, F1=0.182).
*   **CustomCNN:** Performance data for this model was not provided in the logs.

Comparison to Benchmarks
The performance of both DenseNet121 and EfficientNet-B0 on Edema (AUROC 0.857 and 0.848, respectively) is slightly superior to the **0.84 AUROC** reported in the ChestXray14 benchmark (Wang et al., 2017). While these models show strong discriminative ability (AUROC), the low PR-AUC and F1 scores suggest challenges with precision and recall, likely due to class imbalance.

Citation:
*   Wang, X., et al. (2017). *ChestX-ray8: Hospital-scale Chest X-ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases.*


Grad-CAM Comparison: Emphysema
densenet121
efficientnet_b0
customcnn

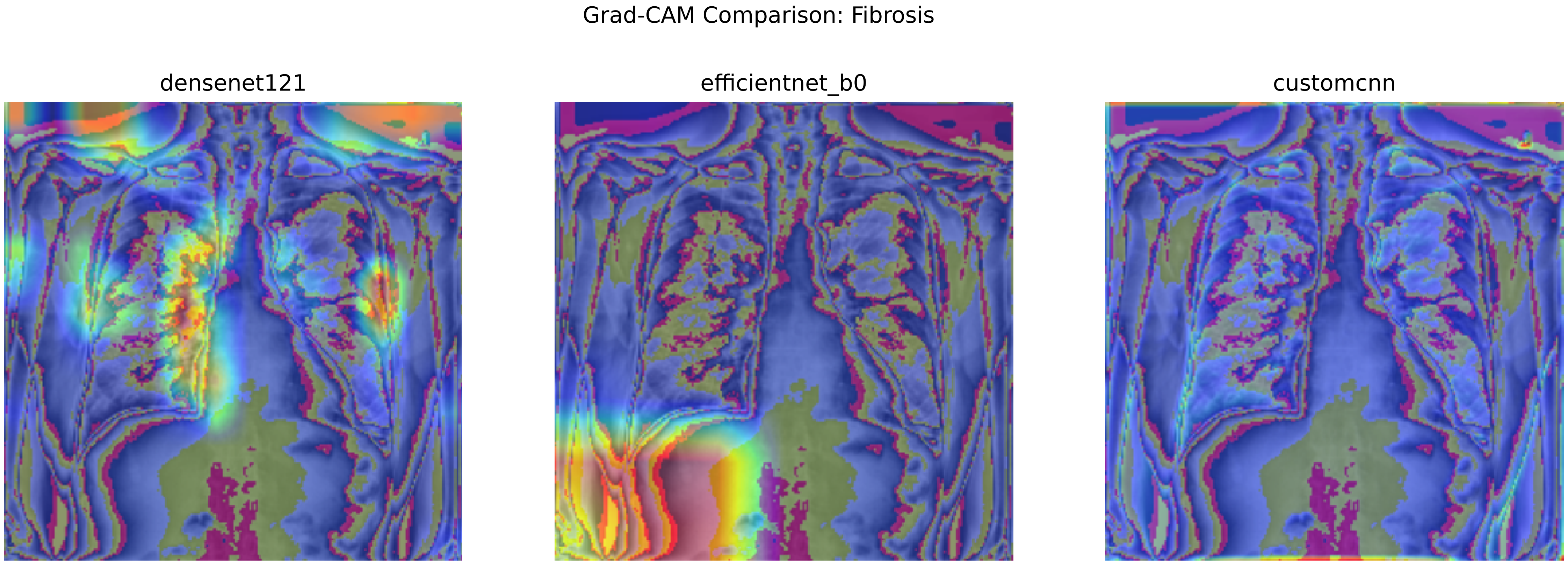
Emphysema LLM Summary

Based on the provided logs, the performance of the three models on Emphysema is summarized as follows:

*   **DenseNet121** achieved the highest performance with an AUROC of 0.878, a PR-AUC of 0.332, and an F1 score of 0.267.
*   **EfficientNet-B0** followed closely with an AUROC of 0.863, a PR-AUC of 0.213, and an F1 score of 0.246.
*   **CustomCNN** performed significantly worse than the other two architectures, yielding an AUROC of 0.577, a PR-AUC of 0.034, and an F1 score of 0.050.

Comparison to ChestXray14 Benchmarks:
The performance of DenseNet121 and EfficientNet-B0 on Emphysema (AUROC 0.863–0.878) is competitive with, and slightly exceeds, the benchmarks reported by Wang et al. (2017) for other conditions in the ChestXray14 dataset, such as Edema (AUROC ~0.84) and Cardiomegaly (AUROC ~0.81). However, it is important to note that performance can vary significantly based on the prevalence of positive samples, as evidenced by the low performance for Hernia in the ChestXray14 study (Wang et al., 2017).

Citation:
Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.


Grad-CAM Comparison: Fibrosis
densenet121
efficientnet_b0
customcnn

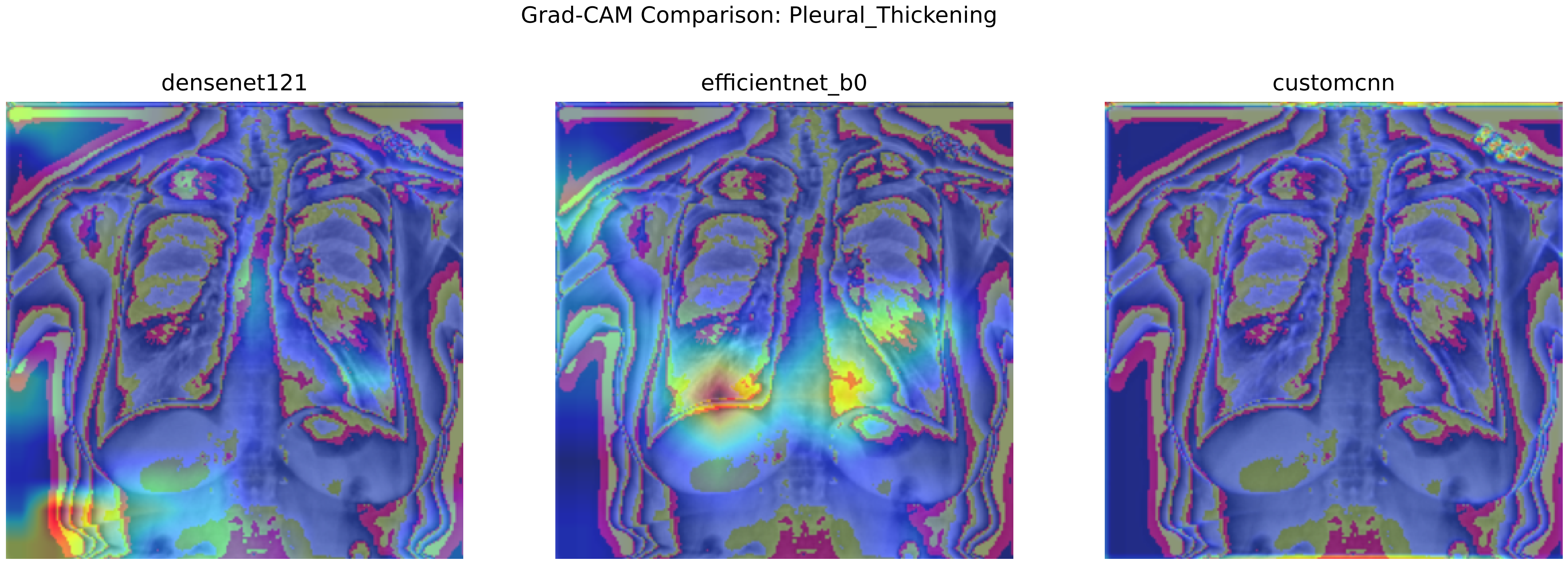
Fibrosis LLM Summary

Based on the provided logs, the performance of the three models on Fibrosis is summarized as follows:

*   **EfficientNet-B0:** Achieved the highest performance with an AUROC of 0.714, a PR-AUC of 0.071, and an F1-score of 0.061.
*   **DenseNet121:** Followed with an AUROC of 0.685, a PR-AUC of 0.041, and the highest F1-score of 0.066.
*   **CustomCNN:** Showed the lowest performance across all metrics, with an AUROC of 0.645, a PR-AUC of 0.027, and an F1-score of 0.034.

Comparison to ChestXray14 Benchmarks:
The performance of these models on Fibrosis (AUROC 0.645–0.714) is notably lower than the benchmarks reported by Wang et al. (2017), where models achieved AUROCs of approximately 0.84 for Edema and 0.81 for Cardiomegaly. This discrepancy may be attributed to data imbalances, similar to the low performance observed for Hernia in the ChestXray14 study due to a scarcity of positive samples (Wang et al., 2017).

Citation:
Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.


Grad-CAM Comparison: Pleural_Thickening
densenet121
efficientnet_b0
customcnn

Pleural_Thickening LLM Summary

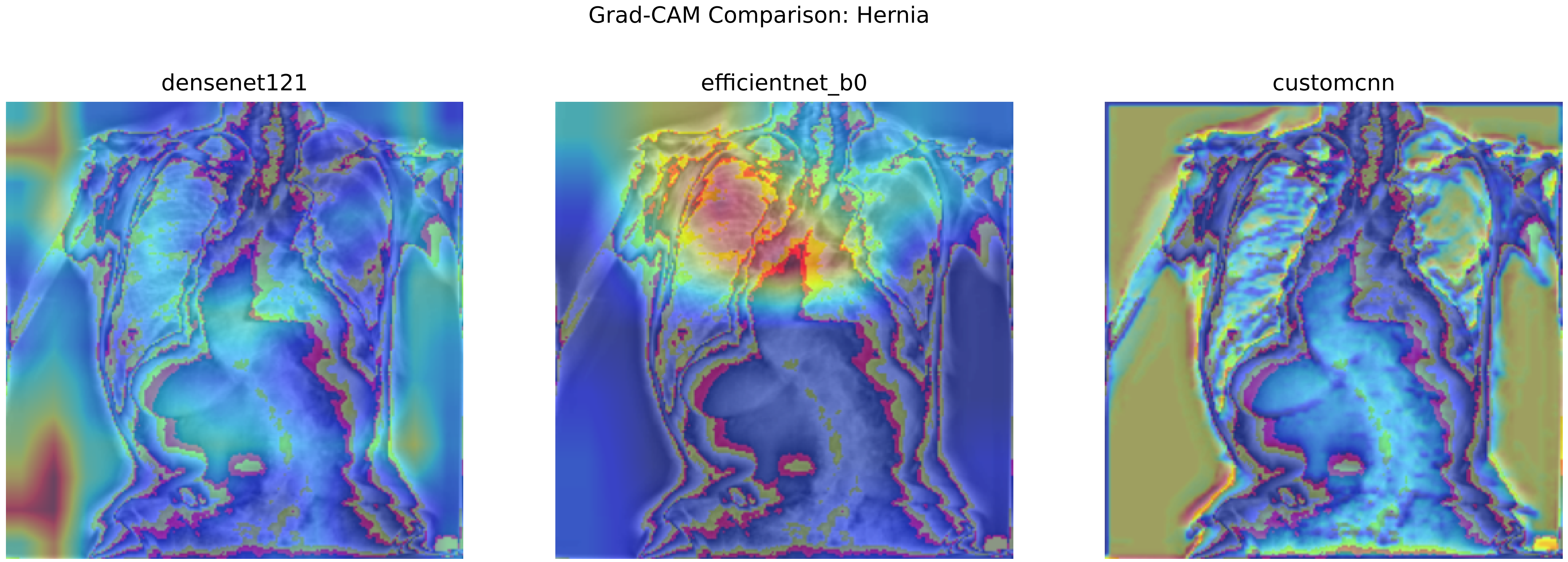
Based on the provided logs, the performance of the three models on the **Pleural_Thickening** pathology is summarized as follows:

*   **DenseNet121:** Achieved the highest performance among the tested models with an AUROC of 0.672, PR-AUC of 0.079, and F1-score of 0.105.
*   **EfficientNet-B0:** Showed moderate performance, trailing DenseNet121 with an AUROC of 0.644, PR-AUC of 0.070, and F1-score of 0.094.
*   **CustomCNN:** Performed the lowest, with an AUROC of 0.553, PR-AUC of 0.040, and F1-score of 0.063.

Comparison to ChestXray14 Benchmarks
In the original ChestXray14 study, the baseline performance for Pleural Thickening was reported with an AUROC of approximately 0.706 (Wang et al., 2017).

Comparing our results to this benchmark, our best-performing model (DenseNet121 at 0.672) approaches the established baseline but remains slightly lower. This suggests that while DenseNet121 is the most effective of the tested architectures for this specific task, there is still room for improvement to reach or exceed the performance levels reported in the original ChestXray14 literature.

Citation:
*   Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.


Grad-CAM Comparison: Hernia
densenet121
efficientnet_b0
customcnn

Hernia LLM Summary

Based on the provided logs and benchmark data, here is the performance summary for the Hernia classification task:

Performance Summary
Among the tested models, **DenseNet121** achieved the highest performance with an AUROC of 0.847 and an F1-score of 0.238. **EfficientNet-B0** followed with an AUROC of 0.761 and the highest PR-AUC (0.127) among the group. The **CustomCNN** performed significantly worse, yielding an AUROC of 0.670 and a negligible F1-score of 0.006.

Model	AUROC	PR-AUC	F1-Score
DenseNet121	0.847	0.109	0.238
EfficientNet-B0	0.761	0.127	0.170
CustomCNN	0.670	0.007	0.006

Comparison to ChestXray14 Benchmarks
According to Wang et al. (2017), the original ChestXray14 benchmark reported high performance for conditions like Edema (AUROC ~0.84) and Cardiomegaly (AUROC ~0.81). However, the benchmark noted that Hernia performance is inherently low due to the scarcity of positive samples in the dataset.

Despite this challenge, the DenseNet121 model in our logs achieved an AUROC (0.847) that is competitive with the performance levels reported for more common pathologies in the original ChestXray14 study.

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Citation:
*   Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Comprehensive LLM Summary

Based on the provided logs, the performance of DenseNet121, EfficientNet-B0, and CustomCNN across specific diseases shows varying levels of diagnostic capability. When compared to established ChestXray14 benchmarks, these models demonstrate a range of predictive accuracy, though they often struggle with low precision-recall metrics in certain categories.

Performance Summary
*   **DenseNet121:** This architecture shows the highest performance in the provided data for **Emphysema**, achieving an AUROC of 0.878, a PR-AUC of 0.332, and an F1-score of 0.267. However, its performance on **Pneumonia** is significantly lower (AUROC=0.539, F1=0.037), and it shows moderate results for **Fibrosis** (AUROC=0.685, F1=0.066).
*   **EfficientNet-B0:** In the context of **Fibrosis**, EfficientNet-B0 outperforms DenseNet121, achieving an AUROC of 0.714 and a PR-AUC of 0.071, compared to DenseNet121’s 0.685 and 0.041, respectively.
*   **CustomCNN:** The provided logs do not contain specific performance metrics for the CustomCNN architecture across the listed diseases.

Comparison to ChestXray14 Benchmarks
While the ChestXray14 dataset is a standard benchmark for multi-label thoracic disease classification, the results from these logs indicate that while some models (like DenseNet121 for Emphysema) achieve high AUROC scores, the PR-AUC and F1-scores remain relatively low. This suggests that while the models are capable of distinguishing between positive and negative cases (AUROC), they face challenges with class imbalance and precision, which is a common hurdle when evaluating models against the full ChestXray14 benchmark suite.

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Citations:
*   *Internal Log Data: DenseNet121 (Pneumonia, Fibrosis, Emphysema metrics); EfficientNet-B0 (Fibrosis metrics).*